

INTRODUCTION

PHYS505 – Electromagnetic Theory

Y. Torun

Illinois Institute of Technology

Maxwell's equations

$$\vec{\nabla} \cdot \vec{D} = \rho$$

$$\vec{\nabla} \times \vec{H} - \frac{\partial \vec{D}}{\partial t} = \vec{J}$$

$$\vec{\nabla} \times \vec{E} + \frac{\partial \vec{B}}{\partial t} = \vec{0}$$

$$\vec{\nabla} \cdot \vec{B} = 0$$

- Set of 1st order linear partial differential equations
- for vector fields $\vec{D}$, $\vec{H}$, $\vec{E}$, $\vec{B}(\vec{r}, t)$
- with sources ρ , $\vec{J}$
- Simplify in vacuum
- Predict charge conservation (continuity equation follows from 1st two)
- and speed of light

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$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}$$

$$\vec{\nabla} \times \vec{B} - \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} = \mu_0 \vec{J}$$

$$\vec{D} = \epsilon_0 \vec{E}$$

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$$\frac{1}{c^2} = \mu_0 \epsilon_0$$

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Exercise

Derive the continuity eqn.

Force and fields

- Lorentz force

$$\vec{F} = q (\vec{E} + \vec{v} \times \vec{B})$$

used to define the fields in original experiments

- force per unit charge for $\vec{E}$
- force per unit current for $\vec{B}$
- but fields are the fundamental quantities in the theory (regardless of whether there are any charges to experience forces),
- Fields can exist in regions with no sources,
- and carry energy, momentum and angular momentum

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ϵ_0 , μ_0 are fudge factors for engineering (SI) units

$$c = 299792458 \text{ m/s}$$

$$\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$$

$$\epsilon_0 = \frac{10^7}{4\pi c^2} \simeq 8.85418781762039 \times 10^{-12} \text{ F/m}$$

$$\frac{10^7}{4\pi\epsilon_0} = 89875517873681764 \text{ (m/s)}^2$$

ϵ_0 , μ_0 are fudge factors for engineering (SI) units (**m**, **kg**, **s**, **A**)

$$c = 299792458 \text{ m/s}$$

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Exercise

Derive the **MKSA** forms of the units for ϵ_0 , μ_0 from Maxwell's eqns.

Why are these people happy?

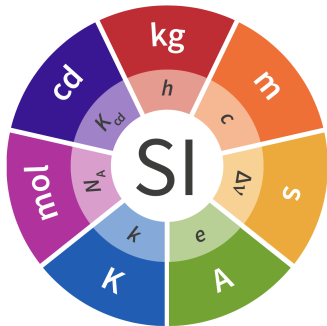


NIST News – Nov 16, 2018: Historic Vote Ties Kilogram and Other Units to Natural Constants



The U.S. delegation at the 26th General Conference of Weights and Measures where more than 55 countries voted to redefine four of the seven base units for the International System of Units (SI).

SI units – time, distance



- Originally (until 1960)

$$s = (\text{Earth day})/86400$$

- Now (since 1967), hyperfine transition in ^{133}Cs

$$\Delta\nu = 9192631770 \text{ s}^{-1}$$

- Originally (1793-1960)

$$m = (1/4 \text{ Earth circumf})/10^7$$

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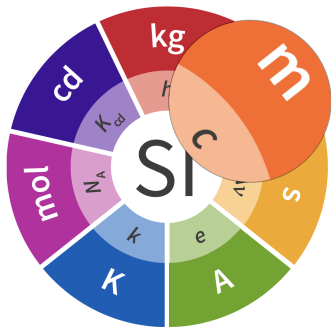
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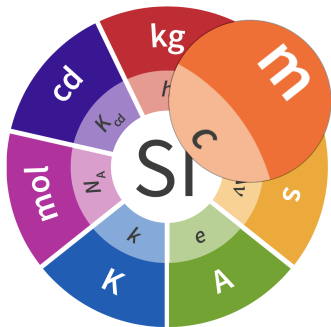
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SI units update – mass, current

- 1889-2018

kg = mass of Le Grand K

- New, Planck constant

$$h = 6.62607015 \times 10^{-34} \text{ kg m}^2/\text{s}$$

- 1946-2019

$$\text{A} \leftrightarrow F = 2 \times 10^{-7} \text{ N/m at } d = 1 \text{ m}$$

- New, electron charge

$$e = 1.602176634 \times 10^{-19} \text{ A s}$$



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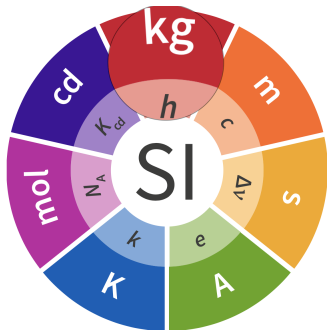
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SI units update – temperature, mole

- 1967-2019

Triple pt of H_2O = 273.16 **K**

- New, Boltzmann constant

$$k = 1.380649 \times 10^{-23} \text{ J/K}$$

- 1967-2019

12 g of ^{12}C = **mol**

- New, Avogadro constant

$$N_A = 6.02214076 \times 10^{23} \text{ mol}^{-1}$$



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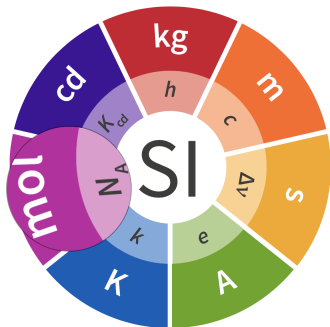
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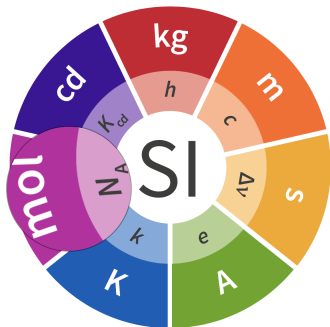
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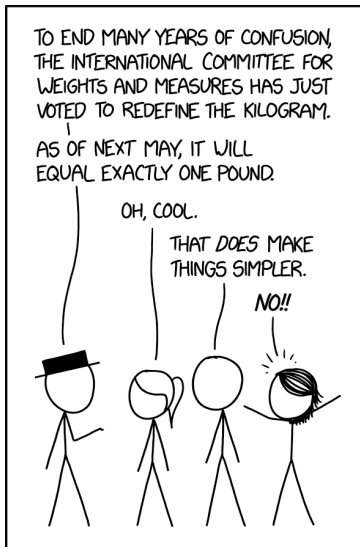
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Tooltip: I'm glad to hear they're finally redefining the meter to be exactly three feet.

Classical vs quantum

- Accurate description of nature requires quantum mechanics
- There is a quantum theory of electromagnetism (Quantum Electrodynamics)
- and a unified description of weak and electromagnetic interactions within the Standard Model
- We will be studying electromagnetism as a classical field theory – the field components are real-valued functions, not operators in a Hilbert space
- Classical electrodynamics is an excellent approximation whenever the number of photons is large while the momentum/energy of individual photons is small compared to the relevant (macroscopic) scale in the system

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- and for point charges, accept arbitrary (non-discrete) values
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Example

A few years ago, I ran an experiment at Fermilab in which we were sending EM waves with frequency 200 MHz and rms power 4 MW over a waveguide with 4" diameter to a radio frequency cavity.

- The average intensity is

$$I = \frac{P}{A} \simeq \frac{4.0 \times 10^6 \text{ W}}{\pi (2 \times 2.54 \times 10^{-2} \text{ m})^2} \simeq 0.49 \text{ GW/m}^2$$

- The peak electric and magnetic field are given by

$$E_0 = \sqrt{2 Z_0 I} \simeq \sqrt{2 (4.9 \times 10^8 \text{ W/m})^2 (377 \Omega)} \simeq 0.61 \text{ MV/m}$$

$$B_0 = \frac{E_0}{c} \simeq \frac{6.1 \times 10^5 \text{ V/m}}{3.00 \times 10^8 \text{ m/s}} \simeq 2.0 \text{ mT}$$

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- The corresponding photon flux is

$$\frac{I}{E_\gamma} = \frac{4.9 \times 10^8 \text{ W/m}^2}{1.33 \times 10^{-25} \text{ J}} \simeq 3.7 \times 10^{33} \text{ photons/s/m}^2$$

- This is a rate of 3×10^{31} photons/s through the waveguide

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Editors' Suggestion

Featured in Physics

Search for Invisible Axion Dark Matter with the Axion Dark Matter Experiment

N. Du,¹ N. Force,¹ R. Khatiwada,¹ E. Lentz,¹ R. Ottens,¹ L. J. Rosenberg,¹ G. Rybka,^{1,*} G. Carosi,² N. Woollett,² D. Bowring,³ A. S. Chou,³ A. Sonnenschein,³ W. Wester,³ C. Boutan,⁴ N. S. Oblath,⁴ R. Bradley,⁵ E. J. Daw,⁶ A. V. Dixit,⁷ J. Clarke,⁸ S. R. O'Kelley,⁸ N. Crisosto,⁹ J. R. Gleason,⁹ S. Jois,⁹ P. Sikivie,⁹ I. Stern,⁹ N. S. Sullivan,⁹ D. B. Tanner,⁹

(ADMX Collaboration) and G. C. Hilton¹⁰

¹University of Washington, Seattle, Washington 98195, USA

²Lawrence Livermore National Laboratory, Livermore, California 94550, USA

³Fermi National Accelerator Laboratory, Batavia, Illinois 60510, USA

⁴Pacific Northwest National Laboratory, Richland, Washington 99354, USA

⁵National Radio Astronomy Observatory, Charlottesville, Virginia 22903, USA

⁶University of Sheffield, Sheffield S3 7RH, United Kingdom

⁷University of Chicago, Illinois 60637, USA

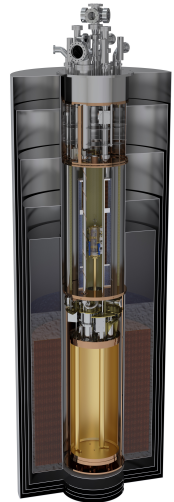
⁸University of California, Berkeley, California 94720, USA

⁹University of Florida, Gainesville, Florida 32611, USA

¹⁰National Institute of Standards and Technology, Boulder, Colorado 80305, USA

 (Received 4 January 2018; published 9 April 2018)

This Letter reports the results from a haloscope search for dark matter axions with masses between 2.66 and 2.81 μeV . The search excludes the range of axion-photon couplings predicted by plausible models of the invisible axion. This unprecedented sensitivity is achieved by operating a large-volume haloscope at subkelvin temperatures, thereby reducing thermal noise as well as the excess noise from the ultralow-noise superconducting quantum interference device amplifier used for the signal power readout. Ongoing searches will provide nearly definitive tests of the invisible axion model over a wide range of axion masses.



Symmetry in Maxwell's eqns

$$\begin{aligned}\vec{\nabla} \cdot \vec{D} &= \rho \\ \vec{\nabla} \cdot \vec{B} &= 0 \\ \vec{\nabla} \times \vec{H} - \partial_t \vec{D} &= \vec{J} \\ \vec{\nabla} \times \vec{E} + \partial_t \vec{B} &= 0\end{aligned}$$

- Consistent with special relativity
- No magnetic charge (monopole) or current
- This is a matter of convention as it's possible to get the same equations through a redefinition of the fields and sources as long as all particles have the same ratio of magnetic to electric charge
- Also $E \leftrightarrow B$ under Lorentz transformations

Inverse square force law

$$F(r) = k q_1 q_2 \frac{1}{r^2}$$

Experimental tests mostly reported two ways

① Assume

$$F(r) \propto \frac{1}{r^{2+\epsilon}}$$

and quote value/upper limit for ϵ

- Cavendish (1772) found $|\epsilon| \leq 0.02$ while Maxwell reported a limit of 5×10^{-5} in similar (concentric spheres) setup

② Assume Yukawa type electrostatic potential

$$V(r) \propto \frac{e^{-\mu r}}{r}$$

with the mass of the photon m_γ given by

$$m_\gamma = \frac{\hbar}{c} \mu$$

and quote value/limit for μ or m_γ

Inverse square force law – Cavendish

221]

THE LAW OF ELECTRIC FORCE.

105

a smaller globe inclosed within it and communicating with the outer one by some conducting substance is rendered at all over or undercharged; and thereby to discover the law of the electric attraction and repulsion.

219] I took a globe 12.1 inches in diameter, and suspended it by a solid stick of glass run through the middle of it as an axis, and covered with sealing-wax to make it a more perfect non-conductor of electricity. I then inclosed this globe between two hollow pasteboard hemispheres, 13.3 inches in diameter, and about $\frac{1}{16}$ of an inch thick, in such manner that there could hardly be less than $\frac{1}{10}$ of an inch distance between the globe and the inner surface of the hemispheres in any part, the two hemispheres being applied to each other so as to form a complete sphere, and the edges made to fit as close as possible, notches being cut in each of them so as to form holes for the stick of glass to pass through.

By this means I had an inner globe included within an hollow globe in such manner that there was no communication by which the electricity could pass from one to the other.

I then made a communication between them by a piece of wire run through one of the hemispheres and touching the inner globe, a piece of silk string being fastened to the end of the wire, by which I could draw it out at pleasure.

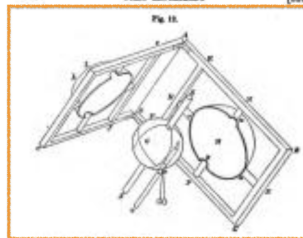
220] Having done this I electrified the hemispheres by means of a wire communicating with the positive side of a Leyden vial, and then, having withdrawn this wire, immediately drew out the wire which made a communication between the inner globe and the outer one, which, as it was drawn away by a silk string, could not discharge the electricity either of the globe or hemispheres. I then instantly separated the two hemispheres, taking care in doing it that they should not touch the inner globe, and applied a pair of small pith balls, suspended by fine linen threads, to the inner globe, to see whether it was at all over or undercharged.

221] For the more convenient performing this operation, I made use of the following apparatus. It is more complicated, indeed, than was necessary, but as the experiment was of great importance to my purpose, I was willing to try it in the most accurate manner.

106

FIRST EXPERIMENT

[381



ABCD and *A'B'C'D'* (Fig. 13) are two frames of wood of the same size and shape, supported by hinges at *A* and *D* in such manner that each frame is movable on the horizontal line *AD* as an axis. *AB* is one of the hemispheres, fastened to the frame *ABCD* by the four sticks of glass *Mm*, *Nn*, *Pp*, and *Rr*, covered with sealing-wax. *A'* is the other hemisphere fastened in the same manner to the frame *A'B'C'D'*. *G* is the inner globe, suspended by the horizontal stick of glass *H*, the frame of wood by which *H* and the hinges at *A* and *D* are supported being not represented in the figure to avoid confusion.

H is a stick of glass with a slip of tinfoil bound round it at *a*, the place where it is intended to touch the globe, and the pith balls are suspended from the tinfoil.

The hemispheres were fixed within their frames in such manner that when the frames were brought near together the edges of the hemispheres touched each other all round as near as might be, so as to form a complete sphere, and so that the inner globe was inclosed within them without anywhere touching them, but on the contrary being at nearly the same distance from them in all parts.

Inverse square force law – Cavendish

230]

ON THE LAW OF ELECTRIC FORCE.

109

229] It must be observed that if the globe was at all overcharged the pith balls should separate further when they were previously positively electrified than when negatively, as in the first case the pith balls must evidently separate further than they would do if the globe was not overcharged, and in the latter case less.

Moreover, a much smaller degree of electricity may be perceived in the globe by this manner of trying the experiment than the former, for when the pith balls have already got a sufficient quantity of electricity in them to make them separate, a sensible difference will be produced in their degree of divergence by the addition of a quantity of fluid several times less than what was necessary to make them separate at first. It is plain that this method of trying the experiment is not just, unless the hemispheres are electrified in nearly the same degree when the pith balls are previously electrified positively as when negatively, which was provided for by the electrometer.

230] In order to find how small a quantity of electricity in the inner globe might have been discovered by this experiment, I took away the hemispheres with their frames, leaving the globe and the pith balls as before. I then took a piece of glass, coated as a Leyden vial, which I knew by experiment contained not more than $\frac{1}{10}$ th of the quantity of redundant fluid on its positive side than the jar by which the hemispheres were electrified did, when both were charged from the same conductor.

I then electrified this coated plate to the same degree, as shewn by the electrometer, that the jar was in the former experiment, and then separated it from the prime conductor, and communicated its electricity to the jar, which was not at all electrified. Consequently the jar contained only $\frac{1}{10}$ th part of the redundant fluid in this experiment that it did in the former, for the coated plate and jar together contained only $\frac{1}{10}$ th, and therefore the jar alone contained only $\frac{1}{10}$ th.

By means of this jar, thus electrified, I electrified the globe in the same manner that the hemispheres were in the former experiment, and immediately after the electrifying wire was withdrawn, approached the pith balls. The result was that by previously electrifying the balls, as in the second way of trying the experi-

110

FIRST EXPERIMENT

[231

ment, the electricity of the globe was very manifest, as the balls separated very sensibly more when they were previously electrified positively than when negatively, but the electricity of the globe was not sufficient to make the balls separate, unless they were previously electrified.

It is plain that the quantity of redundant fluid communicated to the globe in this experiment was less than $\frac{1}{10}$ th part of that communicated to the hemispheres in the former experiment, for if the hemispheres themselves had been electrified they would have received only $\frac{1}{10}$ th of the redundant fluid they did before, and the globe, as being less, received still less electricity.

231] It appears, therefore, that if a globe 12.1 inches in diameter is inclosed within a hollow globe 13.3 inches in diameter, and communicates with it by some conducting substance, and the whole is positively electrified, the quantity of redundant fluid lodged in the inner globe is certainly less than $\frac{1}{10}$ th of that lodged in the outer globe, and that there is no reason to think from any circumstance of the experiment that the inner globe is at all overcharged.

232] Hence it follows that the electric attraction and repulsion must be inversely as the square of the distance, and that when a globe is positively electrified, the redundant fluid in it is lodged intirely on its surface.

For by Prop. V. [Art. 20], if it is according to this law, the whole redundant fluid ought to be lodged on the outer surface of the hemispheres, and the inner globe ought not to be at all over or undercharged, whereas, if it is inversely as some higher power of the distance than the square, the inner globe ought to be in some degree overcharged.

233] For let ADB (Fig. 13) be the hemispheres and adb the inner globe, and Aa the wire by which a communication is made between them. By Lemma IV. [Art. 18], if the electric attraction and repulsion is inversely as some higher power of the distance than the square, the redundant fluid in ADB repels a particle of fluid placed anywhere in the wire Aa towards the center, and consequently, unless the inner globe was sufficiently overcharged to prevent it, some fluid would flow from the hemispheres to the globe.

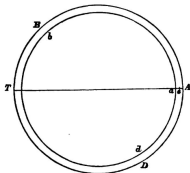
Inverse square force law – Cavendish

234]

ON THE LAW OF ELECTRIC FORCE.

111

Fig. 18.



But if the electric attraction and repulsion is inversely as some lower power of the distance than the square, the redundant fluid in ABD impels the particle in the contrary direction, that is, from the center, and therefore the inner globe must be undercharged.

234] In order to form some estimate how much the law of the electric attraction and repulsion may differ from that of the inverse duplicate ratio of the distances without its having been perceived in this experiment, let AT be a diameter of the two concentric spheres ABD and abd , and let Aa be bisected in e . Ac in this experiment was about $\frac{35}{2}$ of an inch and Tc 13.1 inches, therefore if the electric attraction and repulsion is inversely as the $2 + \frac{1}{10}$ th power of the distance, it may be shewn that the force with which the redundant fluid in ABD repels a particle at e towards the center is to that with which the same quantity of fluid collected in the center would repel it in the contrary direction as 1 to 57.

But as the law of repulsion differs so little from the inverse duplicate ratio, the redundant fluid in the inner globe will repel the point e with very nearly the same force as if it was all collected in the center, and therefore if the redundant fluid in the inner globe is $\frac{1}{57}$ th part of that in ABD the particle at e will be in equilibrium, and as e is placed in the middle between A and a , there is the utmost reason to think that the fluid in the whole wire Aa will be so too. We may therefore conclude that the electric attraction and repulsion must be inversely as some power of the distance between

112

SECOND EXPERIMENT

[235

that of the $2 + \frac{1}{10}$ th and that of the $2 - \frac{1}{10}$ th, and there is no reason to think that it differs at all from the inverse duplicate ratio*.

235] EXPERIMENT II. A similar experiment was tried with a piece of wood 12 inches square and 2 inches thick, inclosed between two wooden drawers each 14 inches square and 2 inches deep on the outside, so as to form together a hollow box 14 inches square and $\frac{1}{2}$ thick, the wood of which it was composed being $\frac{5}{8}$ to $\frac{3}{4}$ of an inch thick.

The experiment was tried in just the same manner as the former. I could not perceive the inner box to be at all over or undercharged, which is a confirmation of what was supposed at the end of Prop. IX. [Art. 41]—that when a body of any shape is overcharged, the redundant fluid is lodged entirely on the surface, supposing the electric attraction and repulsion to be inversely as the square of the distance†.

DEMONSTRATION OF COMPUTATIONS IN [ART. 234].

Let acf be a sphere, c its center, b any point within it, af a diameter, Ee any plane perpendicular to af .

Let $cb = a$, $ba = d$, $bf = s$ and $ad = x$, and let the repulsion be inversely as the n power of the distance. The convex surface of the segment Eae is to that of the whole globe as $ad : af$, and therefore if the point d is supposed to flow towards f , the fluxion of the surface Eae is proportional to x , and the fluxion of its repulsion on b in the direction dc is proportional to

$$\frac{\dot{x}(d-x)}{be^{n+1}},$$

or may be represented thereby, but

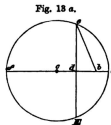
$$be^n = (d-x)^n + x(2a + 2d - x) = d^n + 2ax,$$

therefore the fluxion of the repulsion is

$$\frac{\dot{x}(d-x)}{(d^n + 2ax)^{\frac{n+1}{2}}},$$

* [Note 19.]

† [Art. 561.]



New Experimental Test of Coulomb's Law: A Laboratory Upper Limit on the Photon Rest Mass

E. R. Williams, J. E. Faller, and H. A. Hill

Phys. Rev. Lett. **26**, 721 – Published 22 March 1971

ABSTRACT

A high-frequency test of Coulomb's law is described. The sensitivity of the experiment is given in terms of a finite photon rest mass using the Proca equations. The null result of our measurement expressed in the form of the photon rest mass squared is $\mu^2 = (1.04 \pm 1.2) \times 10^{-19} \text{ cm}^{-2}$. Expressed as a deviation from Coulomb's law of the form $\frac{1}{r^{2+q}}$, our experiment gives $q = (2.7 \pm 3.1) \times 10^{-16}$.

This result extends the validity of Coulomb's law by two orders of magnitude.

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Inverse square force law

Therefore $\vec{E}(r)$ is given by

$$\vec{E}(r) = (qr^{-2} - \frac{1}{3}\mu^2 V_0 e^{i\omega t} r) \hat{r}, \quad (4)$$

where q is the total charge on the inner shell.

A complete solution of the fields inside a symmetrically charged single sphere of radius R_2 gives^{9,10}

$$\begin{aligned} \vec{E}(r) = & \frac{\mu^2}{k^3 r^3} \frac{R_2 V_0 e^{i\omega t}}{e^{-ikR_2} - e^{ikR_2}} \\ & \times [ikr(e^{-ikr} + e^{ikr}) - (e^{-ikr} - e^{ikr})] \hat{r} \end{aligned} \quad (5)$$

and also $\vec{H}=0$. Here $k^2 = \omega^2/c^2 - \mu^2$. A power series expansion of $\vec{E}(r)$ where $kr < 1$ [near zone] and $\omega/c > \mu$ gives

$$\vec{E}(r) = -\frac{1}{3}\mu^2 V_0 e^{i\omega t} r (1 - \frac{1}{10}k^2 r^2 + \frac{1}{6}k^2 R_2^2 \dots) \hat{r}. \quad (6)$$

On neglecting the second-order term, which for this experiment produces an error of less than 1%, this equation reduces to a nonzero rest mass term which is the same as was derived rather simply above.

Since $\partial \vec{H} / \partial t$ is zero inside, $\oint \vec{E} \cdot d\vec{l} = 0$. The voltage appearing across the inductor is then simply given by

$$\int_{R_1}^{R_2} \vec{E} \cdot d\vec{l} = \frac{q}{C} - \mu^2 \frac{V_0 e^{i\omega t}}{6} (R_2^2 - R_1^2). \quad (7)$$

The differential equation which describes this LC circuit in the case of a nonzero rest mass is then

$$L \frac{d^2 q}{dt^2} + r \frac{dq}{dt} + \frac{q}{C} = \mu^2 \frac{[V_0 e^{i\omega t}]}{6} (R_2^2 - R_1^2). \quad (8)$$

The signal-to-noise ratio of a system described by this differential equation can be analyzed using conventional circuit theory. On doing this it turns out that the use of a high frequency, high-

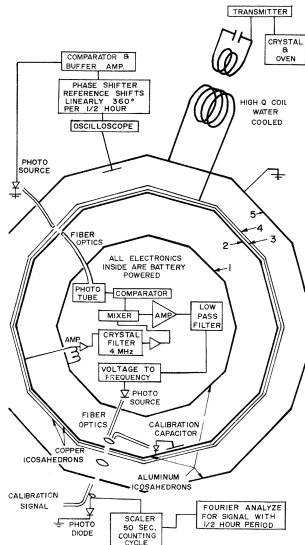


FIG. 1. Schematic drawing of the apparatus. A 4-MHz voltage is applied between shells 4 and 5. A signal of less than 10^{-12} V could have been detected between shells 1 and 2.

NEW GEOMAGNETIC LIMIT ON THE MASS OF THE PHOTON

Alfred S. Goldhaber and Michael Martin Nieto*

Institute for Theoretical Physics, State University of New York at Stony Brook, Stony Brook, New York 11790

(Received 25 June 1968)

On the basis of recent precise data on the earth's magnetic field, a new upper bound on the mass of the photon is obtained using a method that was originally proposed by Schrödinger. The limit is $1.15 \times 10^{-10} \text{ cm}^{-1} \approx 2.3 \times 10^{-15} \text{ eV} \approx 4.0 \times 10^{-48} \text{ g}$, which is four orders of magnitude better than the best laboratory measurement, that of Plimpton and Lawton.

In conventional electrodynamics the mass of the photon is assumed to vanish. However, a finite photon mass may be accommodated in a unique way by changing the inhomogeneous Maxwell equation to the Proca equation,

$$\partial^\lambda F_{\lambda\nu} + \mu^2 A_\nu = (4\pi/c) J_\nu, \quad (1a)$$

where

$$F_{\lambda\nu} = \partial_\lambda A_\nu - \partial_\nu A_\lambda. \quad (1b)$$

If electric charge is conserved, we may rewrite Eq. (1) as

$$(\square + \mu^2) A_\nu = (4\pi/c) J_\nu \quad (2a)$$

where a and b are the radii of the two spheres. The result of Plimpton and Lawton gives

$$\begin{aligned} \mu &\leq 9.8 \times 10^{-7} \text{ cm}^{-1} \\ &\approx 1.9 \times 10^{-11} \text{ eV} \\ &\approx 3.4 \times 10^{-44} \text{ g}. \end{aligned} \quad (6)$$

Another method, first proposed by Schrödinger,² is to use measurements of the earth's magnetic field. From Eq. (3), the dipole ($\vec{m}$) contribution to the earth's field, measured from the center of the dipole, becomes

$$\begin{aligned} \vec{H} = & (e^{-\mu r}/r^3) [(1 + \mu r + \frac{1}{3}\mu^2 r^2) \\ & \times (3\vec{m} \cdot \hat{r} \hat{r} - \vec{m}) - \frac{2}{3}\mu^2 r^2 \vec{m}]. \end{aligned} \quad (7)$$

Using plasma physics to weigh the photon

D D Ryutov

Lawrence Livermore National Laboratory, Livermore, CA 94551, USA

E-mail: ryutov1@llnl.gov

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Abstract

The currently accepted value for the upper bound for the photon mass, m_{ph} , is 22 orders of magnitude less than the electron mass. As the mass m_{ph} is so incredibly small, it has essentially no effect on atomic and nuclear physics; and it is very difficult to improve this estimate by laboratory experiments. However, even a very small mass may have a significant effect on astrophysical phenomena occurring on a scale exceeding the photon Compton length λ (where $\lambda > 3 \text{ Mkm}$ for the currently accepted mass). A set of magnetohydrodynamic equations (assuming a finite photon mass) are used to analyze properties of the solar wind at Pluto's orbit. This yields an improved (reduced) by a factor of 70 estimate of the photon mass. Possible opportunities and challenges for the further reduction of the upper limit for m_{ph} based on the properties of larger-scale astrophysical objects are discussed.

Inverse square law $\Leftrightarrow$ photon mass

1. Introduction

Contrary to sometimes professed views, the finiteness of the photon mass is compatible with Einstein's relativity theory. The quantum description of a finite-mass spin-one particle was developed by Proca [1]; for a modern exposition see, e.g., Landau and Lifshitz [2]. [The appropriately modified Maxwell equations can be found in textbooks, including Jackson's text \[3\].](#) A general discussion of the earlier work on the massive photon has been presented in an excellent review paper by Goldhaber and Nieto [4]; later developments have been discussed in reviews [5–8].

The currently accepted value for the upper bound of the photon mass [9] is impressively low,

$$m_{\text{ph}} < 10^{-22} m_e \approx 10^{-49} \text{ g}, \quad (1)$$

- [1] Proca A 1930 Sur l'équation de Dirac *Compt. Rend.* **190** 1377
- [2] Landau L D and Lifshitz E M 1987 *Quantum Theory of Fields* (Oxford: Pergamon)
- [3] Jackson J D 1975 *Classical Electrodynamics* (Wiley: Chichester)
- [4] Goldhaber A S and Nieto M M 1971 Terrestrial and extraterrestrial limits on the photon mass *Rev. Mod. Phys.* **43** 277
- [5] Byrne J C 1977 Cosmic tests of Maxwell's equations *Astrophys. Space Sci.* **46** 115
- [6] Zellinger A, Weihs G, Jennewein T and Aspelmeyer M 2005 Happy centenary, photon *Nature* **433** 230
- [7] Tu L C, Luo J and Gilles G T 2005 The mass of photon *Rep. Prog. Phys.* **68** 77
- [8] Okun L B 2006 Photon: history, mass, charge *Acta Phys. Pol.* **37** 565
- [9] Eidelman S and Particle Data Group 2004 Review of particle physics *Phys. Lett. B* **592** 1
Yao W-M and Particle Data Group 2006 Review of particle physics *J. Phys. G: Nucl. Part. Phys.* **33** 1

Inverse square law $\Leftrightarrow$ photon mass

Using plasma physics to weigh the photon

B437

Table 1. Some of the available estimates of the photon mass.

References	[15]	[16]	[18]	This paper	[17, 28]	This paper	[4]
Object	Jupiter	Solar wind, 1 AU	Ultra-low frequency waves	Solar wind at 40 AU	Galactic magnetic field	Cluster magnetic field	The Universe
λ (cm, lower bound)	3×10^{10}	3×10^{11}	7.5×10^{11}	2×10^{13}	3×10^{21}	3×10^{23}	2×10^{28}
m_{ph} (g, upper bound)	10^{-48}	$10^{-49\text{a}}$	4×10^{-49}	$1.5 \times 10^{-51\text{b}}$	$10^{-59\text{c}}$	$10^{-61\text{c}}$	$1.5 \times 10^{-66\text{d}}$

^a This is the currently accepted limit.

^b This is the limit suggested in this paper.

^c Insufficient information is available regarding the magnetic field structure in these objects. Accordingly, these values of the photon mass are not quite reliable.

^d This is actually a lower bound on the photon mass, related by the uncertainty principle to the age of the Universe.



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FRB 121102 casts new light on the photon mass



Luca Bonetti^{a,b,c}, John Ellis^{d,e}, Nikolaos E. Mavromatos^{d,e}, Alexander S. Sakharov^{f,g,h},
Edward K. Sarkisyan-Grinbaum^{h,i,*}, Alessandro D.A.M. Spallicci^{a,b,c}

^a Observatoire des Sciences de l'Univers en région Centre, UMS 3116, Université d'Orléans, 1A rue de la Férollerie, 45071 Orléans, France

^b Pôle de Physique, Collegium Sciences et Techniques, Université d'Orléans, Rue de Chartres, 45100 Orléans, France

^c Laboratoire de Physique et Chimie de l'Environnement et de l'Espace, UMR 7328 Centre Nationale de la Recherche Scientifique, LPC2E, Campus CNRS, 3A Avenue de la Recherche Scientifique, 45071 Orléans, France

^d Theoretical Particle Physics and Cosmology Group, Physics Department, King's College London, Strand, London WC2R 2LS, United Kingdom

^e Theoretical Physics Department, CERN, CH-1211 Genève 23, Switzerland

^f Department of Physics, New York University, 4 Washington Place, New York, NY 10003, United States

^g Physics Department, Manhattan College, 4513 Manhattan College Parkway, Riverdale, NY 10471, United States

^h Experimental Physics Department, CERN, CH-1211 Genève 23, Switzerland

ⁱ Department of Physics, The University of Texas at Arlington, 502 Yates Street, Box 19059, Arlington, TX 76019, United States

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ABSTRACT

The photon mass, m_γ , can in principle be constrained using measurements of the dispersion measures (DMs) of fast radio bursts (FRBs), once the FRB redshifts are known. The DM of the repeating FRB 121102 is known to $< 1\%$, a host galaxy has now been identified with high confidence, and its redshift, z , has now been determined with high accuracy: $z = 0.19273(8)$. Taking into account the plasma contributions to the DM from the Intergalactic medium (IGM) and the Milky Way, we use the data on FRB 121102 to derive the constraint $m_\gamma \lesssim 2.2 \times 10^{-14} \text{ eV} c^{-2}$ ($3.9 \times 10^{-50} \text{ kg}$). Since the plasma and photon mass contributions to DMs have different redshift dependences, they could in principle be distinguished by measurements of more FRB redshifts, enabling the sensitivity to m_γ to be improved.



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^a Observatoire des Sciences de l'Univers en région Centre, UMS 3116, Université d'Orléans, 1A rue de la Fêrolierie, 45071 Orléans, France

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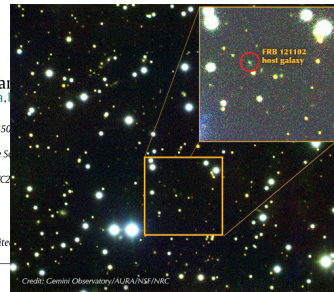
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Superposition

- Maxwell's eqns are *linear* in the fields $\vec{E}$, $\vec{B}$
- which means we can use superposition to solve problems
- Nonlinear effects are present in materials
- but bad things happen on paper even in vacuum
- The electromagnetic energy of a localized charge distribution blows up as the size goes to 0
- $E(r) \rightarrow \infty$ as $r \rightarrow 0$ for a point charge
- Attempts to impose a cutoff/saturation for fields at small distance have failed
- Superposition works in the quantum description of multi-electron atoms
- where fields are of order $1/(4\pi\epsilon_0)e/a_0^2 \approx 10^{12}$ V/m
- The embarrassment persists in the quantum theory (swept under the rug by renormalization)

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- Superposition works in the quantum description of multi-electron atoms
- where fields are of order $1/(4\pi\epsilon_0)e/a_0^2 \approx 10^{12}$ V/m
- The embarrassment persists in the quantum theory (swept under the rug by renormalization)

Superposition

- Maxwell's eqns are *linear* in the fields $\vec{E}$, $\vec{B}$
- which means we can use superposition to solve problems
- Nonlinear effects are present in materials
- but bad things happen on paper even in vacuum
- The electromagnetic energy of a localized charge distribution blows up as the size goes to 0
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Nonlinearity – vacuum polarization

Quantum electrodynamics has corrections for light scattering that amounts to an effective Coulomb potential between point charges $Z_1 e$ and $Z_2 e$ that looks like

$$V(r) = \frac{e^2}{4\pi\epsilon_0} \frac{Z_1 Z_2}{r} \left[1 + \frac{2\alpha}{3\pi} \int_{2m}^{\infty} d\kappa \frac{\sqrt{\kappa^2 - 4m^2}}{\kappa^2} \left(1 + \frac{2m^2}{\kappa^2} \right) e^{-\kappa r} \right]$$

to first order in the fine structure constant $\alpha \simeq 1/137.036$

$$\alpha = \frac{1}{4\pi\epsilon_0} \frac{e^2}{\hbar c} \simeq 0.00729735$$

and m is the inverse Compton wavelength $m = m_e c / \hbar$ implying a short range correction at distances below electron Compton wavelength

$$\lambda_e = \frac{1}{m} = \alpha a_0 \simeq 3.9 \times 10^{-13} \text{ m}$$

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which is negligible for light atoms (orbital radius a_0/Z).

Macroscopic media

- Maxwell's equations specify continuous fields defined over arbitrarily small distances
- We won't attempt to solve them with $\mathcal{O}(10^{24})$ sources (individual atoms)
- We are typically interested in the average of the source and field variables over a large number of microscopic constituents
- referred to as *macroscopic* fields and sources
- equations for macroscopic $\vec{E}$, $\vec{B}$ look the same

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- while $\vec{D}$ and $\vec{H}$ are not necessarily linear in $\vec{E}$ and $\vec{B}$

$$D_{\alpha} = \epsilon_0 E_{\alpha} + \left(P_{\alpha} - \sum_{\beta} \frac{\partial Q'_{\alpha\beta}}{\partial x_{\beta}} + \dots \right)$$
$$H_{\alpha} = \frac{1}{\mu_0} B_{\alpha} - (M_{\alpha} + \dots)$$

- with averaged electric dipole ($\vec{P}$), quadrupole ($Q'_{\alpha\beta}$), magnetic dipole ($\vec{M}$) and higher moment densities that represent the response of the medium in the presence of applied fields

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Macroscopic media

- We need *constitutive relations* for the medium in order to solve Maxwell's eqns

$$\vec{D} = \vec{D}[\vec{E}, \vec{B}]$$

$$\vec{H} = \vec{H}[\vec{E}, \vec{B}]$$

which may not be simple functions and may depend on history (hysteresis)

- For conductive media, there's also a generalized Ohm's law

$$\vec{J} = \vec{J}[\vec{E}, \vec{B}]$$

- Response in most materials dominated by electric and magnetic polarization $\vec{P}$, $\vec{M}$
- These can be derived from the atomic structure of the material (solid state physics), we will treat them as given properties

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Macroscopic media

- Linear response to weak applied field – good approximation in most materials

$$D_{\alpha} = \sum_{\beta} \epsilon_{\alpha\beta} E_{\beta}$$

$$H_{\alpha} = \sum_{\beta} \mu'_{\alpha\beta} B_{\beta}$$

expressed in terms of the electric permittivity (dielectric) and the inverse magnetic permeability tensor

- These depend on molecular/crystalline properties and temperature as well as frequency/wavevector due to contributions from nearby points in space/time

$$D_{\alpha}(\vec{k}, \omega) = \sum_{\beta} \epsilon_{\alpha\beta}(\vec{k}, \omega) E_{\beta}(\vec{k}, \omega)$$

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Macroscopic media – electric field response

$$\vec{D}(\vec{k}, \omega) = \epsilon(\vec{k}, \omega) \vec{E}(\vec{k}, \omega)$$

- For light and lower frequencies ($\lesssim 1$ THz), the spatial part can be considered local (negligible $\vec{k}$ -dependence)
- Frequency dependence for the dielectric constant ϵ/ϵ_0
 - 2-3 for most solids at optical frequencies with only electrons moving appreciably in response to applied fields
 - 2-20 at low enough frequencies ($\lesssim 1$ MHz) with all the charged particles responding
 - higher in materials with permanent molecular dipole moments (eg. ~ 88 for water at 0°C)

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$$\vec{H}(\vec{k}, \omega) = \mu'(\vec{k}, \omega) \vec{B}(\vec{k}, \omega)$$

often expressed as $\vec{B} = \mu \vec{H}$

- Relative permeability μ/μ_0
 - $\lesssim 1$ for *diamagnetic* materials (no net atomic/molecular angular momentum)
 - $\gtrsim 1$ for *paramagnetic* materials (net angular momentum from unpaired electron)
 - can be much larger, $10 - 10^6$, for *ferromagnetic* materials which have a form of paramagnetism driven by interactions between the atomic magnetic dipoles causing them to align in local domains even in the absence of external magnetic field

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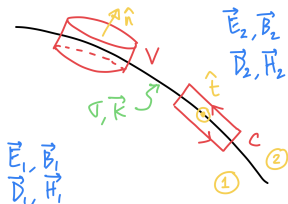
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Boundary conditions at interfaces

- Flux through pillbox with its axis along surface normal $\hat{\mathbf{n}}$
- Line integral over plane curve with its axis along surface tangent $\hat{\mathbf{t}}$
- Sheet of charge (σ), current ($\vec{K}$)



$$\vec{\nabla} \cdot \vec{D} = \rho$$

$$\vec{\nabla} \cdot \vec{B} = 0$$

$$\oint \vec{D} \cdot \hat{\mathbf{n}} da = \int_V \rho d^3x$$

$$\oint \vec{B} \cdot \hat{\mathbf{n}} da = 0$$

$$(\vec{D}_2 - \vec{D}_1) \cdot \hat{\mathbf{n}} = \sigma$$

$$(\vec{B}_2 - \vec{B}_1) \cdot \hat{\mathbf{n}} = 0$$

$$\vec{\nabla} \times \vec{E} = -\partial_t \vec{B}$$

$$\vec{\nabla} \times \vec{H} = \vec{J} + \partial_t \vec{D}$$

$$\oint_C \vec{E} \cdot d\vec{l} = -\int \partial_t \vec{B} \cdot \hat{\mathbf{n}} da$$

$$\oint_C \vec{H} \cdot d\vec{l} = \int (\vec{J} + \partial_t \vec{D}) \cdot \hat{\mathbf{n}} da$$

$$\hat{\mathbf{n}} \times (\vec{E}_2 - \vec{E}_1) = 0$$

$$\hat{\mathbf{n}} \times (\vec{H}_2 - \vec{H}_1) = \vec{K}$$

Idealized constructs and limits

- Fixed potential on a conductor:
 - may require physical connection to allow charge flow
 - wire carrying the current can modify fields in the system
- Surface charge/current density:
 - excess charge/field transition happens within a few lattice constants
 - current confined to macroscopically small, microscopically large thickness (skin depth) in good conductors at high frequency
- Classical electrodynamics works extremely well even at microscopic scales (with atomic sources handled using quantum mechanics)

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